

Recall from earlier, on a given curve $y=f(x)$, the slope of a secant line from $P(a, f(a))$ and $Q(x, f(x))$ is

$$m_{pq} = \frac{f(x) - f(a)}{x - a}$$

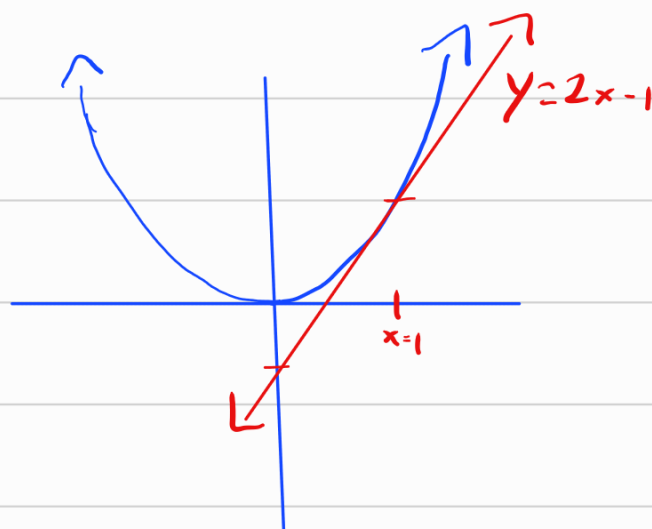
We can then find the slope of the tangent at a by moving Q closer to P . Then the slope is

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

Ex: Tangent line of $f(x) = x^2$ at $a=1$

$$m = \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{(x+1)(x-1)}{x-1} = \lim_{x \rightarrow 1} x+1 = 2$$

Point is $(1, 1)$, tangent line
 $y - 1 = 2(x - 1)$
 $y = 2x - 1$



An equivalent and more common equation for slope at a point is

$$m = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

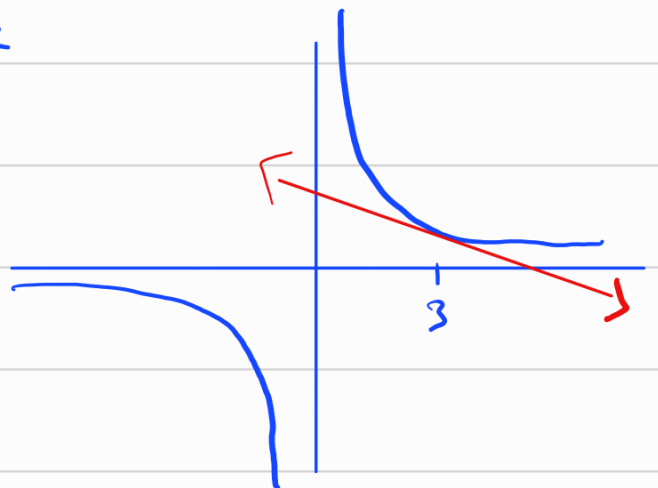
Ex: Tangent line of $f(x) = \frac{3}{x}$ at $a=3$

$$\begin{aligned} m &= \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} = \lim_{h \rightarrow 0} \frac{\frac{3}{3+h} - 1}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{3}{3+h} - \frac{3+h}{3+h}}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{-h}{3+h}}{h} = \\ &= \lim_{h \rightarrow 0} \frac{-1}{3+h} = \boxed{-\frac{1}{3}} \end{aligned}$$

Point is $(3, 1)$, tangent line

$$y - 1 = -\frac{1}{3}(x - 3)$$

$$\boxed{y = -\frac{1}{3}x + 2}$$



Velocities

We discussed earlier that if f is a function that gives position over time, then the instantaneous velocity at point a is just the slope of the tangent at that point. As such,

$$V(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

gives instantaneous velocity at the point a .

Ex: A ball is dropped from a 450m tower.

a) What is velocity after 5 seconds?

b) What is velocity when the ball hits ground?

a) Distance is given by $s(t) = 4.9t^2$

$$\begin{aligned} V(a) &= \lim_{h \rightarrow 0} \frac{s(a+h) - s(a)}{h} = \lim_{h \rightarrow 0} \frac{4.9(a+h)^2 - 4.9a^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{4.9a^2 + 9.8ah + 4.9h^2 - 4.9a^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{9.8ah + 4.9h^2}{h} \\ &= \lim_{h \rightarrow 0} 9.8a + 4.9h = 9.8a \end{aligned}$$

$$s_0, v(a) = 9.8a, \text{ at } a=5, \boxed{v(5) = 49 \text{ m/s}}$$

b) firstly, find the time when ball hits ground.

$$s(t) = 450 = 4.9 t^2$$

$$t = \sqrt{\frac{450}{4.9}} \approx 9.6 \text{ seconds.}$$

$$v(9.6) = 9.8(9.6) \approx 94 \text{ m/s}$$

Derivatives

The formulas we've been working with have a more common name

Def:

The derivative of a function f at a is denoted $f'(a)$ and is given by

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

$$\left(\text{or } \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \right)$$

Ex: $f(x) = x^2 - 8x + 9$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{(a+h)^2 - 8(a+h) + 9 - (a^2 - 8a + 9)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{a^2} + 2ah + \cancel{h^2} - \cancel{8a} - 8h + \cancel{9} - \cancel{a^2} + \cancel{8a} - \cancel{9}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2ah + \cancel{h^2} - 8h}{h} = \lim_{h \rightarrow 0} 2a + h - 8 = 2a - 8$$

$$f'(a) = 2a - 8$$

Ex: $f(x) = \frac{1}{\sqrt{x}}$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a} \frac{\frac{1}{\sqrt{x}} - \frac{1}{\sqrt{a}}}{x - a}$$

$$= \lim_{x \rightarrow a} \frac{\frac{\sqrt{a}}{\sqrt{x}\sqrt{a}} - \frac{\sqrt{x}}{\sqrt{x}\sqrt{a}}}{x - a} = \lim_{x \rightarrow a} \frac{\frac{\sqrt{a} - \sqrt{x}}{\sqrt{x}\sqrt{a}}}{x - a}$$

$$= \lim_{x \rightarrow a} \frac{\sqrt{a} - \sqrt{x}}{(x-a)\sqrt{x}\sqrt{a}} \cdot \frac{(\sqrt{a} + \sqrt{x})}{(\sqrt{a} + \sqrt{x})} = \lim_{x \rightarrow a} \frac{a - x}{(x-a)\sqrt{x}\sqrt{a}(\sqrt{a} + \sqrt{x})}$$

$$= \lim_{x \rightarrow a} \frac{\cancel{-(x-a)}}{(x-a)\sqrt{x}\sqrt{a}(\sqrt{a} + \sqrt{x})} = \lim_{x \rightarrow a} \frac{-1}{\sqrt{x}\sqrt{a}(\sqrt{a} + \sqrt{x})}$$

$$a - x = -(x - a)$$

$$= \frac{-1}{a(2\sqrt{a})} = \frac{-1}{2a^{3/2}}$$

$$f'(a) = \frac{-1}{2a^{3/2}}$$

Ex: $f(x) = \frac{1}{x^2+1}$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{\frac{1}{(a+h)^2+1} - \frac{1}{a^2+1}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{a^2+2ah+h^2+1} - \frac{1}{a^2+1}}{h} = \lim_{h \rightarrow 0} \frac{\frac{a^2+1 - (a^2+2ah+h^2+1)}{(a^2+1)(a^2+2ah+h^2+1)}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-2ah - h^2}{\cancel{h}(a^2+1)(a^2+2ah+h^2+1)}} = \lim_{h \rightarrow 0} \frac{-2a - h}{(a^2+1)(a^2+2ah+h^2+1)}$$

$$= \frac{-2a}{(a^2+1)(a^2+1)} = \frac{-2a}{(a^2+1)^2}$$

$$f'(a) = -\frac{2a}{(a^2+1)^2}$$